

Integral Calculus (2005–2014)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

9

TOTAL MARKS

46 M

TEST DURATION

27 Min

PAPER PATTERN

Classic Paper Pattern (2

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		9	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q27 • ID: JAM_2014_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 The value of $\int_{x=0}^1 \int_{y=0}^{x^2} \int_{z=0}^y (y+2z) dz dy dx$ is

(A) $\frac{1}{53}$

(B) $\frac{2}{21}$

(C) $\frac{1}{6}$

(D) $\frac{5}{3}$

Question 2 (JAM 2013 • Q6 • ID: JAM_2013_Q6)

MCQ

+2 M

Neg: -0.67

The value of the integral

$$\iint_D \sqrt{x^2 + y^2} dx dy, \quad D = \{(x, y) \in \mathbb{R}^2 : x \leq x^2 + y^2 \leq 2x\}$$

is

(A) 0

(B) $\frac{7}{9}$

(C) $\frac{14}{9}$

(D) $\frac{28}{9}$

Question 3 (JAM 2012 • Q4 • ID: JAM_2012_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 The value of $\int_{z=0}^1 \int_{y=0}^z \int_{x=0}^y x y^2 z^3 dx dy dz$ is

(A) $\frac{1}{90}$

(B) $\frac{1}{50}$

(C) $\frac{1}{45}$

(D) $\frac{1}{10}$

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Question 4 (JAM 2011 • Q4 • ID: JAM_2011_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 Let V be the region bounded by the planes $x=0$, $x=2$, $y=0$, $z=0$ and $y+z=1$. Then the value of the integral $\iiint_V y \, dx \, dy \, dz$ is

(A) $\frac{1}{2}$

(B) $\frac{4}{3}$

(C) 1

(D) $\frac{1}{3}$

Question 5 (JAM 2010 • Q2 • ID: JAM_2010_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 The value of $\iint_G \frac{\ln(x^2 + y^2)}{x^2 + y^2} \, dx \, dy$, where $G = \{(x, y) \in \mathbb{R}^2 : 1 \leq x^2 + y^2 \leq e^2\}$, is

(A) π

(B) 2π

(C) 3π

(D) 4π

Question 6 (JAM 2009 • Q8 • ID: JAM_2009_Q8)

MCQ

+6 M

Neg: -2.0

Q.8 Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and $a > 0$. Then the integral $\int_0^a \left[\int_0^x F(y) \, dy \right] dx$ equals

(A) $\int_0^a y F(y) \, dy$

(B) $\int_0^a (a-y) F(y) \, dy$

(C) $\int_0^a (y-a) F(y) \, dy$

(D) $\int_a^0 y F(y) \, dy$

Question 7 (JAM 2008 • Q7 • ID: JAM_2008_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 The set $U = \left\{ x \in \mathbb{R} \mid \sin x = \frac{1}{2} \right\}$ is

(A) open

(B) closed

(C) both open and closed

(D) neither open nor closed

Question 8 (JAM 2005 • Q12 • ID: JAM_2005_Q12)

MCQ

+6 M

Neg: -2.0

12. Let $y = f(x)$ be a smooth curve such that $0 < f(x) < K$ for all $x \in [a, b]$. Let

L = length of the curve between $x = a$ and $x = b$

A = area bounded by the curve, x -axis, and the lines $x = a$ and $x = b$

S = area of the surface generated by revolving the curve about x -axis between $x = a$ and $x = b$

Then

(A) $2\pi KL < S < 2\pi A$

(B) $S \leq 2\pi A < 2\pi KL$

(C) $2\pi A \leq S < 2\pi KL$

(D) $2\pi A < 2\pi KL < S$

Question 9 (JAM 2005 • Q10 • ID: JAM_2005_Q10)

MCQ

+6 M

Neg: -2.0

10. The integral $\int_0^1 \left[\int_0^{1-z} \left(\int_0^2 dx \right) dy \right] dz$ is equal to

(A) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^2 dx \right) dz \right] dy$

(B) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^1 dx \right) dz \right] dy$

(C) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-z} dx \right) dz \right] dy$

(D) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-y} dx \right) dz \right] dy$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Integral Calculus (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	B
2	MCQ	+2	D
3	MCQ	+6	A
4	MCQ	+6	D
5	MCQ	+6	B
6	MCQ	+6	B
7	MCQ	+6	B
8	MCQ	+6	C
9	MCQ	+6	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.